(A) Aggregate kernel statistics at each checkpoint

Statistic	Epoch 10	Epoch 30	Epoch 50	Epoch 70	Epoch 90
Participation ratio	4.90975	5.54409	5.78613	6.02215	5.94795
Global DC energy fraction	0.40941	0.37834	0.36742	0.36042	0.36432
Mean receiver DC fraction	0.48810	0.45970	0.44127	0.43116	0.43160
Signed mass	0.99768	0.99395	0.99248	0.99244	0.99419

(B) Directional change over consecutive 20-epoch intervals

Statistic	Baseline	10 → 30	30 → 50	50 → 70	70 → 90
Mean direction cosine	-	0.536	0.620	0.671	0.722
Mean direction angle (0-180°)	-	55.6°	47.5°	42.1°	36.0°
Orthogonal fraction (0-100%)	-	87.2%	88.8%	89.9%	90.9%